When Delegation Meets Framing: Experimental Evidence on Mass Preferences over Healthcare Spending

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Abstract

Governments often insulate authority over costly and controversial healthcare funding decisions from day-to-day electoral politics, relying on technical assessments and objective criteria to justify those decisions. Yet public responses to these decisions remain highly sensitive to how identical choices are presented. This article examines whether institutional insulation constrains framing effects on mass preferences, or whether citizens' evaluations of state funding decisions remain elastic to framing even when authority is formally insulated from electoral politics. Drawing on prospect theory, we argue that preferences over healthcare spending are reference-point dependent, such that citizens respond differently when funding decisions are framed as avoiding salient losses versus preserving existing gains. We test this argument using a population-based survey experiment embedded in the 2021 Finnish Medicines Barometer, in which respondents evaluate an identical clinical vignette involving a novel, high-cost cancer medicine with uncertain benefits and are randomly assigned to alternative frames emphasizing last-resort rescue or stewardship of finite collective resources. Holding clinical evidence, costs, and uncertainty constant, we find substantial and systematic framing effects on mass preferences over public funding. Support for funding increases when decisions are framed as loss avoidance and decreases when framed as gain preservation, despite identical policy content. These findings demonstrate that institutional insulation from electoral politics does not neutralize framing effects on mass preferences. Instead, decisions justified through technical criteria remain politically consequential because public communication and interpretive frames shape how citizens evaluate distributive choices. By integrating research on framing with the political economy of delegation, this article explains why distributive healthcare policies remain politically contested even when citizens are not the formal decision-makers and highlights the behavioral foundations of public responses to state action under conditions of risk, scarcity, and institutional insulation.

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1 Rationing, delegation, framing and Mass Preferences in healthcare context

All health systems must set priorities in order to allocate scarce fiscal, personnel, and other medical resources. Rationing occurs when potentially beneficial interventions are withheld from individuals (Scheunemann and White 2011). Rationing in healthcare is highly political and is characterised by moral and ethical dilemmas, as well as conflicting values and interests (Baeroe and Baltussen 2014; Baltussen et al. 2016). Mass preferences over high-cost healthcare rationing present a democratic puzzle.

In democracies where funding for healthcare and other public goods is predominantly raised through taxes or tax-like contributions, politics is the main arena for debating the underlying values that guide prioritisation and rationing. However, authority over high-cost and controversial treatment decisions is commonly insulated from day-to-day electoral politics, for example by delegating reimbursement decisions to lower levels of government or to specialised arm's-length agencies that employ applied methods, such as health technology assessment (HTA), to inform decision-making (Egeberg et al. 2009; EC 2018). Nevertheless, rationing decisions always involve value judgements, either explicitly or implicitly embedded in the assessment procedures used (Clark and Weale 2012). For example, HTA frequently includes cost-effectiveness or cost-utility analysis as an evaluation method, the selection of which already reflects a normative commitment to health maximisation as a prioritisation criterion (Jansen et al. 2017; Daniels and van der Wilt 2016).

The replacement or complementing of traditional political institutions by such arrangements could be blamed for blurring democratic control, accountability, and legitimacy (Peters and Pierre 2004). The uptake of innovative pharmacotherapies represents an illustrative example. In Europe, the supranational EU regulatory framework, oriented towards facilitating industrial policy goals (Permanand and Mossialos 2005; Permanand et al. 2006), has led to many new therapies entering markets with limited evidence for their therapeutic value (Hwang et al. 2020; Vella Bonanno et al. 2017). A recent development is the regulation on HTA (EU HTA R (EU)2021/2282), aiming to harmonise and accelerate HTA processes across the member states (Brinkhuis et al. 2024). At national and subnational levels, governments manage the economic risks involved through confidential arrangements with industry (Barrenho and Lopert 2022; Wenzl et al. 2019).

It is widely held that priority setting should be pursued through a framework that engages all relevant stakeholders (Daniels and Sabin 2008; Baeroe and Baltussen 2014; Aidem et al. 2017; Jansen et al. 2018, 68–72). In decisions concerning funding of novel pharmacotherapies, confidentiality, technical complexity, and practical reasons often prevent payers from consulting all stakeholders or disclosing and disseminating the rationales behind decisions to them (Barrenho et al. 2022; Mitton et al. 2006; Kapiriri and Razavi 2017). Thus, decision-making processes recede from the qualities associated with legitimacy (Daniels and Sabin 2008; Daniels and van der Wilt 2016).

Yet decisions are still communicated to and evaluated by citizens. Previous evidence from high-income countries suggests that mass preferences over prioritisation criteria display substantial elasticity in response to how choices are presented and justified. Citizens may not recognise the need to prioritise; they may express conflicting preferences and priorities and value different prioritisation criteria than public payers (Blendon et al. 2012; Owen-Smith et al. 2009; Diederich and Salzmann 2015; Desser et al. 2010; Desser et al. 2013; Ghinea et al. 2021; MacLeod et al. 2016). This elasticity raises an important question: Does institutional delegation insulate these decisions from framing effects, or do citizens' preferences continue to shift with how decisions are framed even when authority is formally insulated from electoral politics?

This article suggests that mass preferences remain elastic to framing effects: preferences over

healthcare funding decisions are reference-point dependent regardless of institutional insulation. Even when clinical evidence, costs, and uncertainty are held constant, citizens' support for expensive medicines changes substantially depending on whether the decision is presented as avoiding a salient loss—withholding a last-resort option from an identifiable patient—or as preserving existing gains by protecting finite fiscal resources from extraordinary expenditure. We find that insulating decision-making from electoral politics does not eliminate the contestability of distributive choices; rather, it reorganizes contestation around public communication and the interpretive frames through which citizens evaluate state action.

Our findings have important implications. Such healthcare policies remain politically contentious not because decisions are justified using allegedly objective technical and scientific criteria rather than political judgment, but because mass preferences remain anchored to loss-versus-gain framing and to rescue narratives that can be activated—or dampened—through how choices are presented to the public. In the pharmaceutical sector, an additional factor is the intervention of well-resourced industry lobbies, capable of deploying costly strategies to pursue and protect their interests (Greer et al. 2016). Conflicts over prioritisation have manifested as critical debates in the media (Abelson et al. 2009; Corbett and Mori 1999), which can be a powerful driver of public opinion (Gabe et al. 2012; Aggarwal et al. 2017; Pinho et al. 2020).

The experiences of the NHS Cancer Drugs Fund in England provide an example of the consequences of such conflicts. The fund was created in response to public and political pressure to finance cancer medicines that the National Institute of Health and Care Excellence (NICE) had not recommended on the basis of its assessment criteria. An evaluation conducted five years after its establishment found no meaningful benefit to patients or society, when measured in terms of survival, quality of life, or toxicity, and suggested potential societal harm once opportunity costs were taken into account (Aggarwal et al. 2017).

Extraordinary costs of new oncology medicines challenge health systems globally. In the Nordic countries, cancer medicine costs increased between 2012 and 2017, with growth rates ranging from 37 to 125 percent, being the fastest-growing cost element of cancer care. Torkki et al. (2022) also document pronounced cross-national differences in cost structures and cost drivers, reflecting variation in the organization of care and health policy priorities. Similar dynamics extend beyond Europe. Using U.S. data, Nguyen et al. (2025) report that very high-cost users tend to live in areas with higher social needs and that conditions such as cancer account for a disproportionate share of expenditures, underscoring the distributive and political dimensions of high-cost care.

These pressures are compounded by scientific uncertainty: novel oncology drugs often enter reimbursement debates with limited evidence on long-term survival or quality-of-life effects, while the fiscal implications of public funding are immediate (Cohen 2017). As Aziz et al. (2020) illustrate, marginal expected health gains are frequently paired with exceptionally high costs. Overall, retrospective analyses of novel cancer medicines approved through accelerated access pathways suggest limited added clinical benefits (Davis et al. 2017; Brinkhuis et al. 2024).

This context generates an important tension between the principles that structure how funding decisions are made and how citizens evaluate these decisions, thus providing substantive terrain on which framing mechanisms operate.[1] One logic emphasizes stewardship: because healthcare budgets

[1]At a structural level, these choices resemble classic problems in distributive politics, where public funds are allocated to projects with concentrated benefits and diffuse costs, such as geographically targeted infrastructure spending (Weingast, Shepsle, and Johnsen 1981). However, healthcare rationing differs in a crucial respect: while residence in a remote island is largely a matter of choice, exposure to severe illness is fundamentally uncertain and broadly shared. As a result, citizens evaluating cancer medicines may support high-cost spending not because they benefit directly today, but because they recognize a latent personal risk of future need—an intertemporal logic that distinguishes healthcare from standard pork-barrel allocations.

are finite, publicly financed systems require allocating resources in ways that sustain broad access and protect population health. The other logic emphasizes rescue: when a specific patient faces imminent death and a treatment offers even a small chance of benefit, refusing to act is experienced as a morally troubling choice. This intuition is captured in the "rule of rescue," which McKie and Richardson (2003, p. 2407) define as the imperative people feel to rescue identifiable individuals facing avoidable death. "This imperative entails a preference for identifiable over statistical lives, the shock-horror response it elicits, and the preference it entails for lifesaving over non-lifesaving measures." In cancer care, these intuitions are repeatedly activated because patients are identifiable, prognoses are often stark, and clinical narratives translate easily into public dramas of avoidable loss. At the same time, these decisions are unavoidably distributive: the opportunity costs of funding one medicine are borne by anonymous others, and citizens must assess whether collective resources have been allocated appropriately.

Existing research indicates that citizens can engage with these trade-offs, yet their expressed preferences are conditional on how decisions are presented and justified. Deliberative exercises suggest that publics can accept scarcity and the need for thresholds under certain procedural conditions. In British Columbia, Bentley et al. (2018) find that participants accepted the principle of resource scarcity and the need for governments to make difficult trade-offs when allocating healthcare resources, and supported the view that cost-benefit thresholds must be set for high-cost drugs. Participants also expected reasonable health benefits in return for large expenditures and supported the view that some drugs do not merit funding. Survey evidence from other contexts reveals similar conditionality. For example, Noh and Son (2025) report that three-quarters of respondents supported reimbursement of high-cost cancer drugs, but that support depended on confidence that drugs were safe, effective, and cost-effective.

However, mass political attitudes under conditions of constraint are highly elastic to how choices are presented. A broad literature on framing demonstrates that preferences over public spending are malleable. In Ghana, Smith et al. (2025) suggest that support for health taxes increased when these taxes were framed as improving public health or promoting fairness. Similarly, Jones (2022) argues that healthcare policy evaluation necessarily depends on which considerations are made salient. Importantly, others suggest that framing alters how risks and outcomes are interpreted. For example, Kahneman and Tversky (1984) suggest that preferences vary markedly when identical outcomes are framed in terms of losses rather than gains. Related work suggests that frames shape mass policy attitudes in other domains, including food regulation (Roh and Niederdeppe 2016) and medical debt (McCabe 2023). During COVID-19, Valenzuela et al. (2021) document how competing frames shaped support for restrictive policies, with exposure to both economic and public health frames reducing support for mobility restrictions.

Taken together, these literatures motivate an interpretation of mass preferences in delegated healthcare rationing that emphasizes reference points, perceived losses, and asymmetric responses to risk. Prospect theory establishes that individuals evaluate choices relative to reference points and exhibit loss aversion, becoming risk-seeking when choices are framed in the loss domain and risk-averse when framed in the gain domain (Kahneman and Tversky 1979; Vis 2011). Applied to healthcare rationing, this predicts that when a contested funding choice is framed as avoiding a salient loss—for example, by emphasizing withholding a last-resort option from an identifiable patient—prospect theory predicts citizens should be more willing to endorse riskier and more expensive options to avert that loss (**Hameleers2020**). Conversely, when the same choice is framed as protecting existing gains—safeguarding finite fiscal resources from extraordinary expenditure—citizens should favor restraint (**Hameleers2020**).

The key question our paper posits is whether these framing effects persist when such decisions are justified through technical criteria and communicated to citizens in contexts where authority

is formally insulated from electoral accountability. Delegation is commonly narrated as a route to depoliticization, in which evidence is assessed through objective criteria, thresholds are applied consistently, and distributive choices are presented as technically grounded rather than politically discretionary (Thatcher and Sweet 2002). In many contemporary healthcare systems, including Finland, citizens are not asked to decide directly on reimbursement but are routinely exposed to funding decisions that are justified in precisely these technical terms, making public interpretation of such decisions politically consequential even under delegation. Yet if mass preferences are systematically sensitive to loss-versus-gain frames (Laenen 2025), then institutional insulation from electoral politics may not eliminate contestation. Instead, contestation may be reorganized around how decisions are communicated and the interpretive frames through which citizens evaluate state action—shifting from political contestation to contestation over meaning and narrative authority.

This implication connects to debates on delegation and accountability, where responsibility for policy outcomes becomes a central object of political contestation. Existing work suggests that delegation is often assumed to allow governments to manage electoral pressures by shifting responsibility for unpopular decisions, but that this effect is conditional. For example, Heinkelmann-Wild et al. (2023) suggest that delegation can facilitate blame avoidance only under specific institutional designs, while Heinkelmann-Wild and Zangl (2020) suggest that the redirection of political pressure depends on whether citizens believe governments can credibly deny responsibility for outcomes. Delegation can therefore redistribute political pressure, but it does not eliminate contestation. In parallel, research on the contestability of expertise highlights that public responses to decisions justified using expert analysis cannot be explained by technical merit alone. As Head (2024) argues, expert policy advice is increasingly challenged by diverse actors, making the interpretation of state action central.

This paper proceeds as follows. section 2 explains our theory, focusing on how prospect theory and the rule of rescue operate in delegated healthcare rationing.

2 Theoretical Framework: Prospect Theory and the Rule of Rescue in Delegated Healthcare

Prospect theory provides the micro-foundations for understanding how citizens evaluate distributive choices in healthcare rationing and why such evaluations are reference-point dependent. Individuals do not assess outcomes in terms of final states, but as gains and losses relative to a reference point (Kahneman and Tversky 1979, p. 263). The canonical prospect-theoretic value function is concave for gains, convex for losses, and steeper in the loss domain (Kahneman and Tversky 1984, p. 342), implying diminishing sensitivity and, crucially, loss aversion: losses loom larger than equivalent gains. These regularities remain among the most robust findings in decision research (Barberis 2013). In the context of healthcare rationing, this implies that the same reimbursement decision can be psychologically represented either as avoiding a loss (not withholding a last-resort option) or as protecting a gain (preserving finite collective resources), and these alternative representations can generate systematically different preferences even when clinical information and costs are held constant.

Reference points are not fixed. They are shaped by expectations and by communicated baselines (Koszegi and Rabin 2006; Koszegi and Rabin 2007). In policy environments, where most citizens encounter specialized decisions through mediated communication, the descriptive frame becomes part of how the baseline is constructed: it indicates what is treated as the default and what is treated as a deviation (Kahneman 2003). Applied to delegated healthcare rationing, citizens may anchor on multiple plausible baselines: existing coverage norms, a generalized expectation of equal

access, an identifiable patient in need, or the collective budget constraint emphasized by expert institutions. The same choice can therefore be interpreted as a departure from an entitlement-like status quo (experienced as a loss) or as a refusal to extend coverage beyond what is fiscally sustainable (experienced as a foregone gain).

Prospect theory also implies a domain-specific pattern of risk-taking. Because the value function is concave over gains and convex over losses, individuals tend to be risk-averse in the gain domain and risk-seeking in the loss domain (the reflection effect) (Kahneman and Tversky 1979; Vis 2011). This matters directly for high-cost oncology medicines, where decisions often involve clinical uncertainty and the possibility of small-probability benefits at high financial cost. When the situation is framed in the loss domain (for example, as denying a last-resort option), citizens should be more willing to tolerate fiscal risk and uncertainty to avert the salient loss (**Hameleers2020**). When the same situation is framed in the gain domain (for example, as protecting finite resources from extraordinary expenditure and preserving broader access), citizens should become more supportive of restraint (**Hameleers2020**). Related mechanisms reinforce this asymmetry. Probability weighting implies that small probabilities can receive disproportionate attention (Kahneman and Tversky 1979; Kahneman and Tversky 1984); in a rescue-oriented loss frame, even a modest chance of benefit can become politically and morally salient relative to diffuse opportunity costs that remain abstract.

These mechanisms should persist under technocratic delegation because frames operate through interpretation rather than through formal institutional responsibility. Delegation is often narrated as depoliticization: evidence is assessed through objective criteria, thresholds are applied consistently, and distributive choices are presented as technically grounded rather than politically discretionary (Thatcher and Sweet 2002). Yet the fact that decisions are justified technically does not immunize publics against reference dependence. Technical criteria themselves must be interpreted, and citizens draw pragmatic inferences about what a communicated rationale implies and what the decision-maker is prioritizing (Flusberg et al. 2024). Moreover, complexity increases reliance on accessible cues. Under uncertainty, citizens frequently lean on heuristics such as availability, which gives greater weight to vivid, easily imagined cases than to abstract aggregates (Tversky and Kahneman 1973). In rationing, the narrative of a specific patient denied a last-resort medicine is cognitively accessible, whereas opportunity costs borne by anonymous others are comparatively opaque. This asymmetry makes it easier for loss frames to dominate public evaluation even when expert institutions supply cost-effectiveness language and probabilistic evidence.

If prospect theory explains why equivalent choices can elicit different preferences depending on the reference point, the rule of rescue explains why certain reference points are uniquely powerful in healthcare. The rule of rescue is the perceived imperative to save identifiable individuals facing avoidable death even when doing so is inefficient and diverts resources from saving more lives elsewhere (McKie and Richardson 2003, p. 2407). It entails a preference for identifiable over statistical lives and a "shock-horror" response that prioritizes lifesaving over non-lifesaving measures (McKie and Richardson 2003, p. 2407). In rationing settings, rescue narratives convert a distributive trade-off into an evaluation of whether the state is allowing an avoidable tragedy to occur. Because rescue frames foreground an identifiable beneficiary and a morally charged counterfactual (a life that could have been saved), they are well suited to placing citizens in the loss domain, thereby activating loss aversion and risk-seeking responses predicted by prospect theory. Extensions also suggest that end-of-life contexts can intensify the perceived moral stakes of denial, including concerns about narrative closure (Sinclair 2022), which further increases the salience of losses relative to diffuse population-level gains.

Taken together, prospect theory and the rule of rescue provide a coherent account of why technocratic delegation should not eliminate framing effects in public evaluations of reimbursement decisions. When an identical reimbursement decision is framed as avoiding a salient loss to

identifiable patients, citizens should be more willing to endorse riskier and more expensive options. When it is framed as preserving collective gains by protecting finite resources, citizens should be more willing to accept restraint. Rescue narratives should magnify the effect of loss framing by intensifying moral salience and by making the focal outcome vivid, concrete, and attributable. These mechanisms operate at the level of cognition and interpretation, and thus remain active even when decisions are formally insulated from day-to-day electoral accountability and justified through expert procedures. In that sense, technocratic delegation can redistribute political pressure without removing contestation: the struggle shifts toward how decisions are communicated, what reference points are made salient, and whether expert-justified restraint is interpreted as legitimate stewardship or as a failure to rescue.

This implication aligns with research on delegation and accountability, which suggests that delegation may facilitate blame avoidance only under specific institutional conditions and only insofar as citizens accept the attributional story that responsibility has shifted (Heinkelmann-Wild et al. 2023; Heinkelmann-Wild and Zangl 2020). It also aligns with work on the contestability of expertise, where public acceptance of expert advice depends on more than technical merit and remains vulnerable to competing narratives (Head 2024). Accordingly, even in delegated settings, mass preferences over high-cost medicines should remain elastic to loss-versus-gain frames (Laenen 2025), and frames that activate rescue and loss aversion should systematically shift support toward more permissive funding positions, while stewardship frames should shift support toward conditionality and rejection.

3 Experimental Evidence on Framing and Mass Preferences in Delegated Healthcare Decisions

The empirical analysis draws on data from the 2021 Finnish Medicines Barometer, a national, cross-sectional population survey administered biennially by the Finnish Medicines Agency to examine experiences, opinions, and values related to health, medicines, and well-being. The 2021 wave included an ad hoc experimental module designed specifically to assess how citizens evaluate expert-justified healthcare funding decisions when identical cases are presented under alternative frames. This design directly reflects the theoretical framework developed above: in contemporary healthcare systems, reimbursement decisions are typically made through expert procedures and justified using technical criteria, yet they remain visible to the public and subject to mass evaluation. The survey questionnaire was drafted with influence from previous studies conducted in other countries, in particular the survey by Blendon et al. (2012), which similarly exposed citizens to real-life decision-making tasks, and Ghinea et al. (2021), which surveyed Australian citizens' attitudes towards public funding of high-cost cancer medicines.

Finland provides an appropriate research context for three reasons. First, the healthcare system is comprehensive and predominantly publicly funded, resembling other advanced welfare states in which questions of healthcare rationing and legitimacy are politically salient. Second, healthcare resource allocation decisions are routinely justified through clinical evidence, cost-effectiveness thresholds, and health technology assessment, making the communicative dimension of these decisions especially important. However, in contrast to many other countries with similar health systems, the justifications for reimbursement decisions of pharmaceuticals are confidential and thus in violation with the transparency criteria of legitimate limit-setting. Third, Nordic healthcare systems face precisely the structural pressures that motivate this study. Technological and demographic change contribute to a widening gap between medical opportunities and fiscal possibilities, which is a common source of controversy among stakeholders.

Data collection was carried out by a professional market research company (Taloustutkimus Ltd) using a pre-recruited online panel of approximately 40,000 Finnish citizens. To reach the target sample size, 10,105 invitations were distributed to panel members. The sample was stratified to ensure balance across gender, age, education level, and geographic region, with eligibility restricted to Finnish-speaking citizens aged 18–79 years. The final analytic sample consists of 2,081 respondents, corresponding to a completion rate of approximately 20.6 percent. While online panels are not probability samples, stratification and quota-based recruitment ensure close correspondence with the Finnish adult population on key sociodemographic dimensions, making the data suitable for inference about mass public opinion in a welfare-state context.

The experimental module employed a randomized between-subjects design focused on framing effects. Respondents were randomly assigned to one of three conditions: a control condition and two treatment conditions that varied only in how the decision context was framed. All respondents received an identical clinical vignette describing a new oncology medicine characterized by high cost, uncertain benefit, and limited patient eligibility (Table 1). The vignette was drafted based on a published assessment by the Council for Choices in Health Care in Finland (COHERE Finland), with similar decision options that are used by their recommendations. COHERE, working in conjunction with the Ministry of Social Affairs and Health, only issues recommendations on including or excluding health technologies in the range of public health services. Decisions on reimbursement for outpatient medicines are conducted by the Pharmaceutical Pricing Board, also working in conjunction with the Ministry of Social Affairs and Health, and for inpatient medicines, the ultimate decision lies with hospitals. The framing manipulation consisted solely of additional information presented between the vignette and the decision task, allowing differences in responses to be attributed to framing rather than to substantive informational differences.

Following the vignette and framing manipulation, respondents were asked which decision regarding the medicine's use they would find acceptable. Response categories captured increasing levels of cost containment: unconditional public funding, conditional funding tied to price reductions, rejection of funding, and an explicit "I don't know" option. These categories map directly onto the theoretical expectations derived from prospect theory, as unconditional funding represents the most risk-acceptant position, conditional funding an intermediate position, and rejection the most risk-averse position.

To estimate the effect of framing on policy preferences, we model respondents' choices using an ordinal logistic regression, reflecting the ordered structure of the outcome variable (Equation 1). The key independent variable is the framing condition, operationalized as indicators for the rule-of-rescue frame and the stewardship (utility-maximizing) frame, with the control condition as the reference category. The specification also includes demographic and socioeconomic covariates commonly associated with healthcare attitudes: gender, age, and income group, as well as indicators capturing health-related exposure, including eligibility for reimbursement of medicines and self-reported medicine expenditures. All models are estimated with survey weights to preserve population representativeness.

Formally, the estimated model can be written as:

$$\Pr(Y_i \leq j) = \text{logit}^{-1}(\tau_j - \beta_1 \,\text{LossRescue}_i - \beta_2 \,\text{GainsStewardship}_i - \boldsymbol{\gamma}\mathbf{X}_i)\,, \tag{1}$$

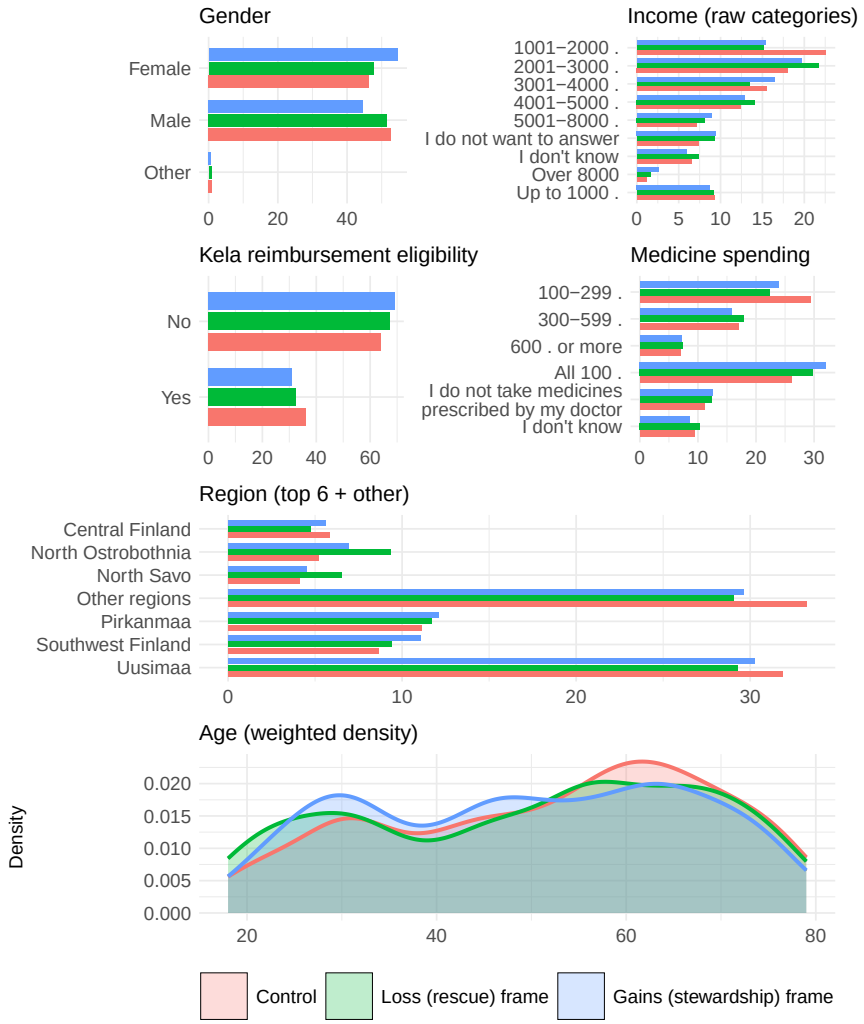
where Y_i denotes respondent i's preferred funding decision, ordered from unconditional funding to rejection; τ_j represents threshold parameters; Rescue_i and Stewardship_i indicate assignment to the loss-framed and gain-framed conditions, respectively; and $\mathbf{X}_i$ is a vector of individual-level covariates including gender, age, income group, health status indicators, and medicine expenditures. Equation 1 therefore estimates how framing shifts the latent propensity to favor more restrictive

Table 1: Decision-making task administered in the survey with framing manipulation.

	Group A (rule-of-rescue frame)	**Group B (health maximisation frame)**	**Group C (control)**
Vignette	Below, you will find a description of the benefits and costs of a new cancer medicine. Please read the description and then answer the question. There are annually 10 to 20 patients in Finland to whom the new medicine can be prescribed. The medicine does not cure cancer but a laboratory study has shown that it destroys cancer cells in about a third of the patients to whom the new medicine is given. It remains unknown if the new medicine extends patients' lives or if it improves their quality of life. The medicine comes with a great risk of adverse effects. If this new medicine is to be approved for use in Finland, its costs will be over 60.000 euros per patient at the final stage of cancer treatment.		
Frame	There is no cure for this particular type of cancer. The new medicine is a possible option for patients who have already received multiple treatments and for whom the remaining options are limited.	The funds available to healthcare are finite. The adoption of the new medicine means that the funds used to pay for it will mean cuts elsewhere in healthcare.	None
Task	What kind of decision regarding the new medicine's use would you find acceptable? Please choose the option that best reflects your opinion. 1. The medicine should be adopted to use and paid by the society regardless of the price set on it by the pharmaceutical company 2. The medicine should be adopted to use and paid by the society if the pharmaceutical company lowers its price 3. The medicine should not be adopted to use and paid by the society 4. I don't know		

funding positions, holding observed characteristics constant.

Figure 1: Frequencies and Densities: Summary Statistics and Covariates Balance



Notes: The figure summarizes sample composition by experimental condition (Control, Loss/rescue, and Gains/stewardship). The age panel displays weighted kernel density estimates of respondents' age. The remaining panels report weighted percentages within each framing condition for the covariates used in the main model (Equation 1). All estimates use survey weights. The close overlap across conditions suggests good covariate balance from random assignment.

3.1 Results

After fitting Equation 1, we show Figure 2 which presents predicted probabilities for the three principal policy preferences across framing conditions (regression table shown in Table A1). The results reveal clear and systematic framing effects. Holding the clinical vignette, cost information, and decision task constant, respondents exposed to alternative frames shifted their preferred policy responses in substantively meaningful ways.

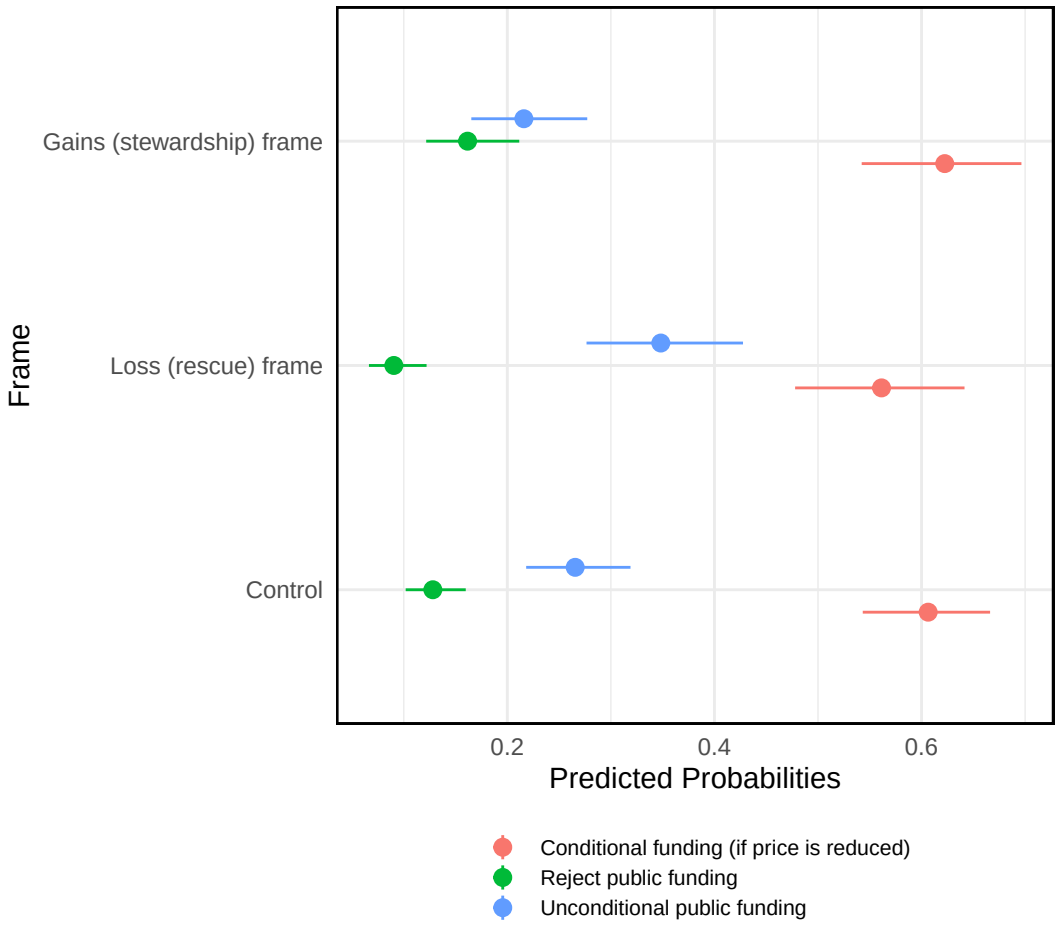


Figure 2: Framing Effects on Public Support for Funding a High-Cost Cancer Medicine

Notes: Points are predicted probabilities from the ordered logistic model (Equation 1; regression table shown in Table A1); vertical bars indicate 90% confidence intervals. The loss (rescue) frame increases support for unconditional funding and reduces outright rejection relative to the control, whereas the gains (stewardship) frame shifts support toward conditional funding and rejection.

The most pronounced pattern concerns support for unconditional public funding, the most risk-seeking option in the decision space. Consistent with the expectations derived from prospect theory, support for unconditional funding increases under the loss frame that emphasizes the absence of alternatives and the proximity of death, and decreases under the gains frame that emphasizes finite budgets and opportunity costs. Conversely, rejection of funding—the most risk-averse response—becomes more likely under the gains frame and less likely under the loss frame. Conditional funding occupies an intermediate position, absorbing shifts in both directions as respondents adjust their evaluations of fiscal risk.

Across conditions, the rank ordering of response options remains stable: unconditional funding is consistently most likely, followed by conditional funding, with outright rejection least likely. What changes is not the rank itself but the level of support attached to each option, as the frames sift probability mass across categories. In particular, the loss frame sifts support upward toward unconditional funding, whereas the gains frame sifts support away from unconditional funding and toward conditional funding and rejection. Thus, framing shifts the level of support across options

while preserving the same ordinal structure of preferences.

These results suggest that citizens do not evaluate reimbursement decisions solely in terms of technical information about clinical effectiveness or cost. Instead, preferences vary systematically with the reference point activated by communication (frames). When the decision is presented as avoiding a salient loss to identifiable patients, respondents appear more willing to accept financial risk and clinical uncertainty. When the same decision is presented as protecting existing resources for the broader patient population, respondents become more cautious and more supportive of restraint. The observed shifts are therefore consistent with the reflection effect and loss aversion predicted by prospect theory, as well as with the moral psychology captured by the rule of rescue.

Importantly, these effects emerge in a setting where respondents are not asked to act as policymakers but to evaluate a decision justified through information resembling real reimbursement debates. This feature of the design speaks directly to the institutional argument advanced in this article. The findings suggest that insulating reimbursement decisions from day-to-day electoral politics and grounding them in technical criteria does not eliminate the role of framing in shaping public evaluation. Citizens remain responsive to how choices are narrated, and their preferences display substantial elasticity even when the informational content of the decision remains constant.

We believe this elasticity has important implications for the legitimacy and contestability of healthcare rationing. Delegation to expert bodies may reduce direct political responsibility for specific reimbursement decisions, but it does not remove those decisions from the arena of public judgment. Instead, it shifts the locus of contestation toward communication, interpretation, and narrative framing. Public reactions to expert-justified decisions depend not only on the underlying evidence but also on whether the decision is understood as prudent stewardship of collective resources or as a morally troubling failure to rescue.

Taken together, the results provide empirical support for the theoretical claim developed in the previous section: reference-point–dependent evaluation and rescue-oriented reasoning persist even under institutional arrangements designed to depoliticize distributive choices. In contexts characterized by uncertainty, high stakes, and competing moral logics, framing remains a central mechanism through which citizens interpret and evaluate state action.

3.2 Discussion

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4 Appendix

4.1 Regression Model

Table A1: Framing Effects on Public Support for Funding Expensive Cancer Medicines

	Ordinal logit
Loss (rescue) frame (vs. control)	1.478***
	[1.252, 1.746]
Gains (stewardship) frame (vs. control)	0.762**
	[0.644, 0.901]
Male (vs. female)	0.953
	[0.830, 1.095]
Age	1.016***
	[1.012, 1.021]
Income: middle (vs. low)	0.858
	[0.719, 1.023]
Income: high (vs. low)	0.944
	[0.778, 1.146]
Income: other/unknown	0.821
	[0.652, 1.035]
Eligible for Kela reimbursement	0.977
	[0.829, 1.151]
Num.Obs.	2458

Notes: Entries report odds ratios from an ordinal logistic regression. The dependent variable measures respondents' preferred funding decision for a novel, high-cost cancer medicine, ordered from unconditional public funding to outright rejection. The key independent variable captures experimental framing of the decision context. Control variables include gender, age, income group, education, eligibility for Kela medicine reimbursement, and self-reported medicine expenditure. Models are estimated using survey weights. 90% confidence intervals in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.